

Integrals – 2

Symmetry, the king's rule and periodicity

1 сентября 2026 года

Theoretical minimum.

Def. The devices of this sheet compute no antiderivative at all: they use the fact that a definite integral is unchanged by substitutions mapping the interval onto itself.

Theorem 1 (the king's rule). $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$. Consequently $\int_a^b \frac{f(x)}{f(x)+f(a+b-x)} dx = \frac{b-a}{2}$, and $\int_0^\pi x f(\sin x) dx = \frac{\pi}{2} \int_0^\pi f(\sin x) dx$.

Theorem 2 (parity and periodicity). The integral of an odd function over a symmetric interval vanishes; for a function of period T the integral over any interval of length T is the same and $\int_0^{nT} f = n \int_0^T f$. Finally, for an even f , $\int_{-a}^a \frac{f(x)}{1+e^x} dx = \int_0^a f(x) dx$.

Fact 1. The substitution $x \mapsto 1/x$ maps $(0, 1)$ onto $(1, \infty)$; hence $\int_0^\infty \frac{\ln x}{1+x^2} dx = 0$ and $\int_0^\infty \frac{dx}{(1+x^2)(1+x^a)} = \frac{\pi}{4}$ for every a .

1. Prove the king's rule and both consequences in Theorem 1. Apply them to $\int_0^\pi \frac{x \sin x}{1+\cos^2 x} dx$ and to $\int_1^{\sqrt{3}} \frac{\arctan x + \operatorname{arccot} x}{x} dx$ (Integration Bee 2022).

2. (Integration Bee) Integrals that collapse by symmetry:

- a) $\int_0^{2\pi} \arccos(\sin x) dx$ (2024);
- b) $\int_0^{2\pi} \tan(\cos x) dx$ (2019);
- c) $\int_0^{2\pi} \sin(\sin x - x) dx$ (2020);
- d) $\int_0^\pi \cos(x + \cos x) dx$ (2022).

In each case state explicitly which substitution and which symmetry you are using.

3. (Integration Bee) Periodicity and the integer part:

a) $\int_0^{2018\pi} |\sin(2018x)| dx$ (2018);

b) $\int_0^{256} (x - \lfloor x \rfloor)^2 dx$ (2013);

c) $\int_0^{2022} (x^2 - \lfloor x \rfloor \lceil x \rceil) dx$ (2022);

d) $\int_{50}^{100} \lfloor \log_2 x \rfloor dx$ (2016).

4. (Integration Bee) Orthogonality of the trigonometric system: $\int_{-2\pi}^{2\pi} (\cos 3x + \sin 2x)(-\sin 2019x + \cos 3x) dx$ (2019), $\int_{-\pi/2}^{\pi/2} \cos(20x) \sin(25x) dx$ (2025), $\int_{-\pi}^{\pi} \sin^{2025} x \cos^{2026} x dx$ (2026). State a general criterion: when is such an integral automatically zero?

5. The substitution $x \mapsto 1/x$: prove both identities of Fact 1 and then compute $\int_0^\infty \frac{\ln x}{1+x^2} dx$, $\int_0^\infty \frac{\ln x}{(1+x^2)^2} dx$ and $\int_0^\infty \frac{dx}{(1+x^2)(1+x^\pi)}$.